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Computational Brain, Creative Mind: Intellectual Freedom as the Upper Limit on Artificial Intelligence

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Abstract

Some generative linguists have long maintained that human beings exhibit a species-specific form of *intellectual freedom* expressible through natural language. With roots in Descartes' effort to distinguish humans from machines, it is the human *use* of their linguistic capacity that demonstrates a unique intellectual and creative potential. This "creative aspect of language use," or CALU, is typically conceived as *neither* determined *nor* random *yet* appropriate to the circumstances of its use. CALU is therefore taken to be beyond the scope of computation. This paper argues that there is substance to this claim and that CALU is likely an unreachable threshold for artificial intelligence (AI) developed via computational means. Background is provided by distinguishing the Cartesians' original "language test" from Turing's later "Imitation Game" for machine intelligence. Moreover, it is shown how Turing's work on computability theory allowed generative linguists to reformulate the original Cartesian problem of language use in the twentieth century. The paper proceeds to show why generative Large Language Models do not reproduce CALU. Then, it argues that characteristics relevant to human intelligence such *agency* and *embodiedness*—while conceivable in future AI systems via computation—are insufficient for attaining CALU. Language use's *non-teleological* nature, in contrast to these characteristics, poses a formidable challenge to AI via computation. Finally, clarifying remarks are offered on the problems posed by testing for CALU, with contrasts to Turing's Imitation Game and tests designed in its lineage.

Keywords: Artificial Intelligence; Computation; Creative Aspect of Language Use; Free Will; Generative Linguistics; Intellectual Freedom

1. Introduction

In December 2024, Rouleau & Levin issued a warning: “The era of machines as deterministic, predictable, and boring objects of fixed structure is coming to an end” (Rouleau & Levin, 2024, 1425). Debates over the distinctions or overlap between engineered machines and biological (“real”) beings now require terminological clarity and humility on the necessary and sufficient conditions for consciousness (mind) and the continuum of “unconventional cognitive systems” (Rouleau & Levin, 2024, 1424). Artificial Intelligence (AI) was naturally implicated.

The fundamental problem is not new: Descartes (2009/1637) sought to distinguish between humans and machines by positing a test by which one could detect the presence of a human soul, or *mind*, unattainable by machines. No machine would be capable, he argued, of replicating the ordinary use of human *language* in its unlimited expressiveness. The apparent linguistic fluency of Transformer-based Large Language Models (LLMs) indeed appears to raise Descartes’ problem anew (e.g., Rees, 2022; Birch, 2024, 300).

This paper develops and defends a variant of Descartes’ problem of language use in contemporary cognitive science and philosophy of mind and applies it to fundamental problems in AI. Specifically, following the generative linguistic (and later “biolinguistic”) approach, Descartes’ problem is taken to be a characterization of species-specific *intellectual freedom* owing to the biological properties of the human mind. This is termed the “creative aspect of language use,” or CALU (Chomsky, 2009a). It is specified according to three dimensions: *stimulus-freedom*, *unboundedness*, and *appropriateness to circumstance*. In short, humans possess a distinctive ability to engage in the unbounded and appropriate expression of thought through natural language at will; the free deployment of cognitive resources, a likely pillar of humanity’s creative and intellectual heights.

We thus ask: does Descartes’ problem remain a challenge for today’s machines, such as LLMs? More importantly, can CALU be specified computationally in future AI systems? Or is CALU the threshold which AI via computation cannot meet?

This paper proceeds to argue that CALU is likely an unreachable limit for AI systems constructed via computational means. The argument is one of graded confidence, not decisiveness, as the possibility is not ruled out in principle. Some scholarly precedent exists for this claim. For example, being *neither* determined *nor* random *yet* appropriate, Baker (2007) previously argued that CALU seems to exceed computational bounds.

This paper specifically argues that LLMs do not replicate CALU. More importantly, characteristics bound up in human intelligence such as *agency* and *embodiedness*—while conceivable in future AI systems via computation—would be insufficient for the artificial reproduction of CALU. In contrast, CALU’s *non-teleological* nature appears to be distinct from these characteristics. This argument intersects with a discussion of Turing’s (1950) Imitation Game and the relative *nature* and *stringency* of the original Cartesian language test relative to tests in Turing’s lineage.

The generative linguistic angle may seem odd given the debate currently underway as to the linguistic competencies of LLMs (e.g., Contreras Kallens et al., 2023; Katzir, 2023; Piantadosi, 2024). Yet, its inclusion bears on the maturity of this idea of intellectual freedom in generative linguistics relative to its scarce attention in other approaches as well as its relevance to modern computing. If this paper's thesis proves controversial, it will not owe to scholars' time-worn familiarity with it.

The paper begins by assessing the association between intellectual freedom and machine intelligence through Turing's late-career scholarship, including his conversation-focused Imitation Game. It then moves to the Cartesian language test as a precursor example of a test for mind intimately associated with intellectual freedom and the divergences between the Cartesian and Turing tests. An underappreciated link between Turing's work and how generative linguists reformulated the Cartesian problem (CALU) in the twentieth century allows for a full explication and defense of the idea. It then turns to why LLMs do not replicate CALU and why CALU is likely an unreachable threshold for AI via computation. The paper closes with comments on fundamental problems testing for CALU in contrast to the Imitation Game and tests designed in Turing's lineage.

2. Intellectual Freedom and Machine Intelligence

Turing's late-career work is a natural starting point. Specifically, his direct and indirect comments on *free will* serve as a springboard from which to explore human intellectual freedom and machine intelligence. Importantly, Turing's views are highly idiosyncratic and contested, and his famous 1950 'Computing Machinery and Intelligence,' may have served practical, in addition to intellectual, purposes (see, Gonçalves, 2023a). That said, his comments offer directional clues about freedom and intelligence.

A brief illustration of the Imitation Game is in order. Turing's (1950) Imitation Game was designed as a test of machine intelligence in which a machine and a human engaged in anonymized conversation with a human interrogator must fool the interrogator into believing they are communicating with another person. Should the interrogator judge the machine's responses a certain percentage of the time to be human, it passes the test.

The Imitation Game was meant to replace the question, 'Can machines think?' (Turing, 1950, 434). Turing predicts that digital computers will be capable, by century's end, of playing the game well enough that an average human interrogator "will not have more than 70 per cent. chance of making the right identification after five minutes of questioning" (Turing, 1950, 442). In the same passage, Turing claims that the "original question, 'Can machines think?' I believe to be too meaningless to deserve discussion" (Turing, 1950, 442). To use the language of 'thinking machines,' Turing (1950, 442) suggests, will be considered so reasonable as to make quibbling about the rigor of its use a mistake; such use will come to be normal.

On free will, Turing briefly notes that an "interesting variant on the idea of a digital computer is a 'digital computer with a random element'" whose randomness is equivalent to the throw of a die (Turing, 1950, 438). He observes: "Sometimes such a machine is described as having free will (though I would not use this phrase myself)" (Turing, 1950, 438). Turing's qualification

(“though I would not use this phrase myself”) could indicate either that Turing believed randomness was *insufficient* for free will or that Turing objects to the *term* “free will” just as he does “think,” as Copeland (2004, 477) explains.

Still, its mere invocation hints at a wider debate. In a May 1951 lecture entitled, ‘Can Digital Computers Think?’, Turing recognized a tension between the *determinacy* of a digital computer and the *apparent freedom* of the human brain: if the behavior of the brain appears to involve free will, but a digital computer—owing to its programming—is “completely determined,” then “[t]hese two facts must somehow be reconciled” (Turing, 1951, 5). He suggests two possibilities: (1) Free will is an illusion; or (2) Free will is real but impossible to detect in behavior. He asks in this vein whether the brain’s behavior is amenable, in principle, to prediction via calculation, noting that some physicists rule this out on the basis of quantum indeterminacy (Turing, 1951, 5).

Turing does not provide his own view. Notably, however, he links the Imitation Game’s emphasis on *appearance*—a machine’s ability to appear intelligent—to free will itself, noting that what matters is *the appearance of randomness* to those unfamiliar with a machine’s construction. He suggests a computer could be built with a “roulette wheel” that provides the appearance of random behavior and, therefore, the appearance of free will to those unfamiliar with its construction (Turing, 1951, 5; see also, Copeland, 2004, 479).¹

The problem of intellectual freedom was perhaps more central to Turing’s work during this period than his explicit references to freedom indicate, as evidenced by his responses to critics (on this point, see, Proudfoot, 2017, 314-315). Geoffrey Jefferson, one such critic, expounded on Descartes’ language test and natural language’s expressiveness as evidence that a being possesses a mind capable of free will in his 1949 ‘Mind of Mechanical Man’ (Jefferson, 1949, 1106-1107). Shortly thereafter, Jefferson writes:

It can be urged, and it is cogent argument against the machine, that it can answer only problems given to it, and, furthermore, that the method it employs is one prearranged by its operator. The “facilities” are provided and can be arranged in any order by “programming” without rebuilding (Jefferson, 1949, 1109).

He continues:

It is not enough, therefore, to build a machine that could use words (if that were possible), it would have to be able to create concepts to *find for itself* suitable words in which to express additions to knowledge that it brought about (Jefferson, 1949, 1110) (emphasis in original).

A machine, lacking freedom, can only devote its resources to those problems which have been set before it by its programmers—and the mere output of words is insufficient to qualify as such freedom.

¹ Copeland (2004, 478) also notes that Turing’s close friend Max Newman said Turing firmly believed the brain in fact possesses a roulette wheel.

Lady Lovelace similarly noted that the

Analytical Engine has no pretensions whatever to *originate* anything. It can do whatever *we know how to order it* to perform. It can *follow* analysis; but it has no power of *anticipating* any analytical relations or truths. Its province is to assist us in making *available* what we are already acquainted with (Lovelace, 1843, 722) (emphases in original).

The Analytical Engine (in Turing's (1950, 450) view, a universal digital computer) was characterized as the follow-up to a human antecedent; humans provide the intellectual direction which it may follow but the Analytical Engine never creates its own knowledge.

Interestingly, Turing interprets Jefferson's remarks in 'Mind of Mechanical Man' narrowly as the *argument from consciousness*. On "the most extreme form of this view," Turing notes, "the only way by which one could be sure that a machine thinks is to be the machine and to feel oneself thinking" (Turing, 1950, 446), a position he deems "the solipsist point of view" and claims "I am sure that Professor Jefferson does not wish to adopt the extreme and solipsist point of view" (Turing, 1950, 446).

In response to Lovelace, Turing notes that although it is true a machine "can only do what we *do* order it to perform," should the machine's program result "in its doing something interesting which we had not anticipated I should be inclined to say that the machine *had* originated something, rather than to claim...that the originality lies entirely with us" (Turing, 1951, 6).

Neither of these responses explicitly address free will, though the absence of the term should not distract us. Proudfoot (2017) argues that the emphasis Turing places on *appearance* in the Imitation Game—and, more specifically, the *response* that such appearance elicits in observers—led him to adopt a compatibilist position on free will.² Indeed, Turing's response to Lovelace in part depends on a human's response to a machine's output—human surprisal. The compatibilist approach attributed to Turing by Proudfoot is *response-dependent*: "x has *free will* (or acts *voluntarily*) if and only if, in actual normal conditions, x appears to have free will to actual normal subjects" (Proudfoot, 2017, 317). On this view, Turing ultimately seemed to hold that *both* free will is possible in humans and machines *and* that behavior is determined—a form of compatibilism (Proudfoot, 2017, 319).

2.1 Turing and the Cartesian Language Test

Whatever Turing's views, intellectual freedom was intimately bound up in constructing a test that assesses a machine's intelligence. The association has historical roots going back to at least Descartes, who put forward a language test for the possession of mind to distinguish between humans and mechanical beings, partly the focus of Jefferson's (1949) work to which Turing responded.

² Proudfoot suggests that Turing's (1951) invocation of a random element is a *rejection* of randomness as a basis for free will given his previously quoted remark in 1950 ("though I would not use this phrase myself"), and the ease with which he defaults to the *appearance* of randomness as a suitable basis is telling (Proudfoot, 2017, 313).

Descartes' contribution originated within the context of the "mechanical philosophy" which held that the natural world was an elaborate construct, designed by a "master craftsman" (Chomsky, 2007, 39). All natural phenomena, on this view, could be explained according to the physical parts that comprise them and the interactions between them (Cohen, 1985, 154) (e.g., a retina is stimulated by direct contact with the light from the surface of a chair (Chomsky, 2007, 41)).

The problem thus arises for Descartes in his 1637 *Discourse on Method*: how can a machine merely constructed to *resemble* a human be distinguished from an actual human? How, that is, can the *mind*—the soul—be identified in another being that looks like us if, as the mechanical philosophy holds, natural phenomena can be accounted in their entirety in terms of physical parts and contact? What evidence points us to the *freedom* enabled by the human's (as he posited) immaterial soul, or mind?

Descartes happily informs us that "if there were machines bearing the image of our bodies, and capable of imitating our actions as far as it is morally possible, there would still remain two most certain tests whereby to know that they were not therefore really men" (Descartes, 2009/1637, 71-72). We are interested here in the first, typically referred to as the "language test" (see, Gunderson, 1964, 197-201). The language test is as follows:

Of these the first is that they could never use words or other signs arranged in such a manner as is *competent to us* in order to *declare our thoughts* to others: for we may easily conceive a machine to be so constructed that it emits vocables, and even that it emits some correspondent to the action upon it of external objects which cause a change in its organs...but not that it should *arrange them variously* so as *appositely to reply to what is said in its presence*, as men of the *lowest grade of intellect can do* (Descartes, 2009/1637, 72) (emphases added).

Descartes provides several criteria that signify possession of a mind: it speaks or otherwise makes signs *intelligibly* ("competent to us"); it does so in order to *express its thoughts* (to "declare" them); it *combines and re-combines* words in an *appropriate* fashion given the context ("arrange them variously" and "appositely to reply"); and finally, its use of language in this fashion is entirely *ordinary* and not a feature of general intelligence (as those of the "lowest grade of intellect can do"). It is *not* sufficient for possession of a mind if a subject: merely outputs words ("emits vocables"); or if it merely outputs words in direct contact with an external force ("correspondent to the action upon it").

In constructing a test for machine intelligence centered on *conversation*, one can see how Turing's Imitation Game may have been influenced by Descartes' language test. Indeed, Dennett (1985, 123-124) suggested as much. Abramson (2011, 548-549) later offered empirical support for this contention: that Turing's notes showed line-colorings and annotations on Jefferson's 'Mind of Mechanical Man,' and that Turing subsequently responded in 1950, strongly suggests Turing's familiarity with the test if not, as Abramson argues, influenced by the content.

3. Three Divergences in the Cartesian and Turing Tests

Despite the similarities, however, there are at least three divergences between how Turing presented his Imitation Game (and how it has been received in popular, and to some extent, academic culture) and the Cartesian test.

First, the Cartesian test was intimately bound up in the limits of an *explanatory account* of the human mind and body. Unable to account for phenomena like ordinary language use through the mechanical philosophy, a “second substance” is postulated (Chomsky, 2006, 5-6). How to *detect* a mind was linked to how the mind is *situated* in this explanatory framework.

Second, and directly related to the first, the emphasis that Turing placed on the *appearance* of intelligent behavior and free will—merely fooling a human interrogator—marked a disjuncture from the Cartesian tradition. The Cartesians were not interested in constructing tests that merely allow one to *speak* of possession of mind or intelligence obtained through demonstrations like ordinary language use. To the contrary, as Chomsky argues:

Though similar in some ways to Turing’s imitation game, the Cartesian tests for other minds are posed within an entirely different framework. These tests are ordinary science, designed to determine whether some object has a particular property, rather like a litmus test for acidity (Chomsky, 2009c, 105).

A major difference between Turing and the Cartesians was the latter’s definite interest in constructing a crucial experiment for the possession of mind; the detection of an independent property in the world.³

Turing’s ambition with the Imitation Game is thus lower than the Cartesians’ language test. For the Cartesians, ordinary language use was indeed “only one illustration of the general problem of will...” (Chomsky, 2009b, 177). It is the non-mechanical nature of this behavior that lends to its status in signifying possession of a mind like our own. The mind, on this view, is a distinct property. The language test is a means of detecting this property. That we today make no appeal to the mechanical philosophy or a second substance does nothing, however, to eliminate the *problem* of ordinary language use—the problem is as real today as it was then. Turing’s Imitation Game moves *away* from the perniciousness of this as a scientific problem by attempting a circumvention. This was evident in Turing’s (perhaps dismissive) response to Jefferson.

Note that this divergence would hold even if Proudfoot’s notion of response-dependence reflected Turing’s beliefs. Response-dependent free will, while closer in a sense to the detection of a real property in the world sought by the Cartesians, is too underspecified to firmly locate it within this tradition.⁴ Moreover, it would lead Turing to have adopted a type of compatibilism in which determinism is compatible with free will. Yet, the Cartesian language test—illustrated in

³ This is consistent with Gonçalves’ interpretation of Turing who argues that the Imitation Game’s perceived status as a “crucial experiment to determine the existence of an intelligent machine” (Gonçalves, 2023a, 360) may have been a “concession to meet the standards of his interlocutors more than this own” (Gonçalves, 2023a, 362), perhaps using the Imitation Game as a tool of rhetoric to advance the *actual building* of an intelligent machine (Gonçalves, 2023a, 382-384).

⁴ To be sure, Gonçalves (2023b) argues that Turing’s Imitation Game, conceived as a thought experiment, does possess a rich grounding in modern scientific tradition. Even here, however, the divergence from the Cartesian project appears to hold.

more detail below—is premised on the idea that ordinary language use is *neither* determined *nor* random (yet appropriate to situations). It is not compatibilist.⁵ The divergence is serious.

Finally, the Cartesian test is considerably more stringent than both Turing’s original formulation of the Imitation Game and commonplace interpretations thereafter. This in part owes to the greater specification of the Cartesian test following Descartes’ initial remarks. It also owes to the characteristic it aims to identify—human will—being less amenable to tests designed in the Imitation Game’s lineage, explored in detail below.

To be sure, as Dennett (1985, 124) explains, Turing’s original test was “severe” in its stringency, particularly insofar as the test’s content was to be “unrestricted” (Dennett, 1985, 131) (any question, on any topic, is permissible). It is therefore possible that perceptions of Turing’s test have weakened its stringency over time in some corners.

Still, Cartesians following Descartes took pains to extend his observations about ordinary language use beyond his brief remarks in the *Discourse on Method*. Géraud de Cordemoy, in his 1668 *Philosophical Discourse Concerning Speech*, takes up the Cartesian association between words and thought in the effort to distinguish bodies that outwardly appear human from those with souls (i.e., minds) (Cordemoy, 1668, 2-6). How, though, does one know that the words uttered through speech reflect a “soul” like one’s own and are not merely a result of “*Mechanicks* one may so fitly adjust certain Bodies to one another” (Cordemoy, 1668, 8)?

Though words and thoughts are linked in humans, to probe for the existence of a mind in bodies that appear human-like it is not sufficient that they exhibit “*the facilness of pronouncing Words*” to conclude “*that they had the advantage of being united to Souls*” (Cordemoy, 1668, 13-14). Diversity of word arrangement is insufficient for possession of mind. Thus, it is not the mere output of words that matters for the distinction between humans and mere machines. Rather, it is how words and other signs are *used*:

But yet, when I shall see, that those Bodies shall make signes, that shall have *no respect at all to the state they are in*, nor to their conversation: when I shall see, that those signs shall agree with those which I shall have made to *express my thoughts*: When I shall see, that they shall *give me Idea’s, I had not before*, and which shall relate to the thing, I had already in my mind: Lastly, when I shall see a *great sequel between their signes and mine*, I shall not be reasonable, If I believe not, that they are such, as I am (Cordemoy, 1668, 18-19) (emphases added).

It is reasonable to accept that the subject possesses a soul/mind if: their use of words lacks a *necessary* or otherwise *fixed connection* with their current state or surroundings (“no respect at all to the state they are in, nor to their conversation”); their words correspond to the *meanings* of the words used by the interlocutor to convey the contents of their mind (“those signs shall agree with those which I shall have made to express my thoughts”); their words provide *novel ideas*

⁵ McGilvray (1979, 538) points out that not all rationalists saw determinism (in the form of control by external stimuli) as incompatible with free will. Nevertheless, the position explicated here cannot be compatibilist because the unboundedness of speech and its stimulus-freedom are unpredictable and undetermined and therefore *libertarian*.

which the interlocutor did not previously possess but which relate to their thoughts (“give me Idea’s, I had not before”); finally, there is a *complementarity* between the use of words by the subject and the interlocutor (“a great sequel”).

Thus, for the Cartesians, the problem of language use is the problem of how humans ordinarily convey the contents of their minds to others by communicating new ideas through novel utterances that correspond with their thoughts, done with no apparent fixed relationship with one’s inner physiological state nor local context.

3.1 The Biolinguistic Reformulation

Though underappreciated in AI discourse today, Turing’s work on general computability theory in the early twentieth century intersected with the Cartesian problem by allowing linguists to *reformulate* the problem. The introduction of computability theory allowed generative linguists in the mid-twentieth century to reformulate the problem of ordinary language use into two separate but overlapping problems: one is the target of linguistic inquiry in the generative, and later biolinguistic, paradigm; the other is left aside.

This is frequently misunderstood. Chomsky’s 1964 *Current Issues in Linguistic Theory* is often quoted, where he says:

The central fact to which any significant linguistic theory must address itself is this: a mature speaker can produce a new sentence of his language on the appropriate occasion, and other speakers can understand it immediately, though it is equally new to them...In fact, the realization that this “creative” aspect of language is its essential characteristic can be traced back at least to the seventeenth century...In any event, whatever the antiquity of this insight may be, it is clear that a theory of language that neglects this “creative” aspect of language is of only marginal interest (Chomsky, 1964, 7-8).

Chomsky is describing two problems: one is the problem of how human beings *use* language to express new thoughts appropriately (i.e., CALU); the other is the *capacity* for infinite generativity from finite means (the mechanism enabling them to generate an unbounded array of structured expressions from finite elements).

The Cartesians made a distinction between body and soul, but no serious inroads could be made within the mechanical philosophy on the infinite generativity from finite elements. It offered no resources to account for how a finite, physical object (the human body) could yield infinite outputs (an unlimited variety of structured linguistic expressions).

It was not until work in the formal sciences on decidability, completeness, and mathematical truth—including by Turing and Alonzo Church, among others—in the early twentieth century that the problem of infinite generativity from finite means could be taken up in earnest. Chomsky indeed used the terms *recursive* and *computable* equivalently in the 1950s, inspired by work in mathematical logic at the time that likely had broad influence (see, Lobina, 2014, 62-63; see also, Mendívil-Giró, 2018, 874). With these new intellectual tools, a generative procedure internal to the human mind could be postulated; “a finite object that characterizes an infinite array of ‘free expressions,’ each a mental structure with a certain form and meaning” (Chomsky,

2007, 45). The generative procedure is *computational*⁶ (Chomsky, 2007, 45-46; see also, Chomsky, 2017, 1-2).⁷

Yet, a generative procedure does not resolve the original Cartesian formulation of the problem of creative language *use*. It targets only the “infinite scope of finite means” (Chomsky, 2007, 46). That is, it targets only what linguists today refer to as *linguistic competence*, or the abstract knowledge of language that underwrites its use. *Linguistic performance*, actual language use, is explicitly left aside (by generativists). Linguistic competence is the target of inquiry; linguistic performance is segmented off from inquiry.

The Cartesian problem of ordinary language use was therefore split into two, reformulated along lines that make its enabling capacity (competence) amenable to scientific inquiry. Indeed, as Murphy notes, biolinguistics merely “seeks to give a naturalistic account of linguistic aspects of thought” without denying the free expression of thought through language (Murphy, 2012, 116).

We should be clear about what exactly this reformulation of the Cartesian problem into competence and performance entails, as it is crucial: computability theory is an intellectual tool that allows linguists to study the properties of a single system capable of producing an infinite array of structured linguistic expressions.⁸ This, in generative linguistics, is a system of internalized knowledge.

This intellectual tool, however, comes with a corollary: a system of internalized knowledge is distinct from the *use* of this system in production, i.e., the actual use of language. Computability theory thus provides for the distinction to be made between the internal competence and externalized performance. *But it crucially has nothing to say about ordinary language use*; the original Cartesian problem (see, Chomsky, 2017, 2). The innovation of the mid-twentieth century was to distinguish “the internalized system of knowledge from the processes that access it” (Chomsky, 2017, 2), thereby making a distinction between internal language and speech (or signs) that the Cartesians appear not to have made for they lacked the tools to make it.

4. The Free Exercise of the Human Intellect

CALU’s role in generative and biolinguistics has remained steady over the years. For our purposes, it is its intimate link with explanation and what constitutes a science of mind that is of interest. Like the Cartesians, CALU in biolinguistics is situated within an explanatory framework (and reformulated in doing so). Chomsky and philosophers of language and mind have, in this vein, taken pains to provide greater specification of what exactly constitutes CALU.

In contemporary terms, CALU is explicated according to three criteria: *stimulus-freedom*, *unboundedness*, and *appropriateness to situations*. Importantly, it is only *together* that the three

⁶ Universal Grammar is the study of this procedure’s innate basis.

⁷ As Millière (2024, 2) notes, cognitive science in its early days was generally inspired by computer science concepts, including the idea that the human mind’s cognition could be explained as computations over structured symbolic representations. That said, our focus here is a result of the relative maturity of this line of Cartesian thought in the generative enterprise.

⁸ Note that one need not accept a domain-specific or domain-general view of language on this matter. The basic point applies regardless. The “single system” could conceivably be either the entire brain or some sub-component(s).

properties constitute the creative aspect of language use. In isolation, they would be insufficient to hold the significance imbued by the original Cartesian problem (Baker, 2007, 236-237). As Chomsky summarizes:

In short, then, man has a species-specific capacity, a unique type of intellectual organization which cannot be attributed to peripheral organs or related to general intelligence and which manifests itself in what we may refer to as the “creative aspect” of ordinary language use—its property being both unbounded in scope and stimulus-free. Thus Descartes maintains that language is available for the free expression of thought or for appropriate response in any new context and is undetermined by any fixed association of utterances to external stimuli or physiological states (identifiable in any noncircular fashion) (Chomsky, 2009a, 60).

CALU is the at will expression of thought through an unbounded range of utterances, lacking any identifiable one-to-one relationship between stimulus and utterance, yet selecting from an unbounded class of possible expressions an utterance that is nonetheless appropriate to the situation at hand and coherent to others (who readily impose an appropriate interpretation on the utterance). CALU importantly does not appear to vary in any discernable measure between individuals. It is instead reflective of an “intellectual organization” of the human mind; its fundamental architecture permits it. Human language use “is not a series of random utterances but fits the situation that evokes it but does not cause it, a crucial if obscure difference” (Chomsky, 1988, 5). It is *neither* determined (stimulus-free and unbounded) *nor* random (appropriate).

To say that utterances are stimulus-free is not to deny internal or external influences. We are deeply inclined to respond to remarks directed at us. Yet, this amounts to linguistic productions being *elicited* by such stimuli, not *caused* by them. The distinction between elicit and cause is nicely illustrated with contrast to animal communication systems, with the latter being “purely functional and stimulus-bound” (Chomsky, 2009a, 63), confined to *local contexts* (Anderson, 2008, 796-797).

Consider Berthet et al.’s (2025) argument that extensive compositionality—the “nontrivial” ability to combine elements with independent meanings into new combinations in which these elements modify one another—is present in the bonobo vocal system. Their argument explicitly uses a framework that defines the meaning of a given bonobo signal “as the set of features of circumstances (FoCs) that appear at a rate greater than chance across the signal’s occurrences” (Berthet et al., 2025, 104). Bonobo ‘utterances,’ then, are explicitly tied to the circumstances of their use. In sharp contrast, human language use is detachable, readily so, from local contexts or even *possible* contexts; it is stimulus-free.

To say that language use is unbounded is to highlight its *infinite productivity* (Huybregts, 2019, 2). Operating over a finite set of elements, humans use the syntax of their language to combine and re-combine them into new forms that convey new meanings (Moro, 2016, 21-23); “an infinite range of discrete, different messages” (Anderson, 2008, 797). The turn to computability theory in linguistics is relevant here: there is an unbridgeable gap between a combinatorial ability

of some finite kind and the *infinite* productivity of human language; its unboundedness. Hence, the need for computability theory. If language is conceived as a computable function (in Turing's (1937) sense), a generative procedure that operates internally, then "[i]nfinite productivity is the default property..." (Huybregts, 2019, 2).

It is conceivable that an organism could possess a linguistic competence capable of this infinite productivity but remain *stimulus-bound*. The unboundedness of language use could be recruited in saying, upon viewing a painting, either "*Dutch*" or "*Clashes with the wallpaper*" (Chomsky, 1959, 52), or any other combination of words. But this is importantly distinct from the *stimulus-freedom* of such utterances. The painting, for this hypothetical organism, could be a fixed point of association with its linguistic production. Conversely, language use could conceivably be detachable from local context but restricted to a finite list of messages. Human language use happens to be *both* stimulus-free and unbounded (the human may indeed say nothing when looking at the painting).

That language use is stimulus-free tells us something about the third dimension: its appropriateness to situations. If language use were stimulus-bound, then it would be a function of stimuli in one's internal or external environments; a matter of either causality or some other fixed association between utterance and stimulus (see, McGilvray, 2001, 8-10). Yet, human language use is not functional; it is *stimulus-free*. More than this, that language use is bound up in the expression of thought means that what counts as "appropriate" is judged by others "who might have reacted in similar ways and whose thoughts, evoked by this discourse, correspond to those of the speaker" (Chomsky, 1988, 5). It is therefore judged "by the much more obscure relation of appropriateness" (Chomsky, 1975, 302).

The detachability of an unbounded range of meaningful expressions from local contexts, which then converge with thoughts of others in what can productively be called a "meeting of minds" (Collins, 2006, 489), is likely a pillar of humanity's intellectual prowess. It is the architecture of the human mind—a mind that enables intellectual freedom—that in part allows the species to reach its creative heights; to escape the strictures of context and immediacy by willfully deploying cognitive resources to problems and human concerns that cross time and space and to mutually contemplate possibilities and impossibilities alike.

4.1 CALU and the Limits of Explanation

CALU finds no apparent scientific explanation. This is unsurprising given that the idea is historically linked with the *limits* of explanation. Computability theory, as noted, did not resolve this problem but instead allowed researchers to reformulate it.

A commonsense view of scientific explanation might hold that language use is beholden to causal factors either in one's *internal* or *external* environment; the key is isolating said factors. Indeed, Chomsky's (1959) critique of Skinner as the representative voice of behaviorism in part hinged on the following counter-argument: that Skinner's (1957) concept of *stimulus control*—a product of behaviorists' orientation to studying "observables" in an organism's environment as the locus of their behavior (Boeckx, 2010, 16-17)—stretched the meaning of the term "stimulus" beyond its scientific usefulness. Moreover, it amounts to post-hoc attribution: "We cannot predict

verbal behavior,” Chomsky wrote, “in terms of the stimuli in the speaker’s environment, since we do not know what the current stimuli are until he responds” (Chomsky, 1959, 32).

Indeed, by attempting to re-trace each linguistic utterance back to a physical object in the individual’s environment, “the word *stimulus* has lost all objectivity in this usage. Stimuli are no longer part of the outside physical world; they are driven back into the organism” (Chomsky, 1959, 52). McGilvray explains that such an attempt at explaining language use “is not serious causality. It is interpretation of an event as part of a pattern” of the given situation (McGilvray, 2001, 7); a post-hoc attribution at best (see also, Asoulin, 2013, 230).⁹

The shift to the study of the mind—to unobservable mental processes and cognitive structures—seems to shift the explanatory burden, then, from the behaviorist’s external causal factors to *internal causal factors*. Put another way: this shift seems to entail that language use is not causally tied to environmental stimuli, but to internal stimuli; the brain *causes* language use.

Yet, as Collins (2006) points out, an *inverse* of Skinner’s problem arises: whether a purported stimulus is internal or external, there is no way of determining this *independently* of an utterance. Causation cannot be established in either direction. In principle, such language use could be explained. This possibility, however, does not aid us if that explanation exceeds the scope of scientific inquiry carried out by humans. Humans seem to lack the science necessary to penetrate an otherwise “identifiable phenomenon” (Collins, 2006, 474).

Surely, though, this merely amounts to an admission of ignorance about the brain’s role in speech behavior, does it not?

One could widen the notion of *causality* to explain language use. Indeed, Potter & Mitchell (2025) make this case for neuroscience. They argue that the field tends to assume that behavior and cognition are underpinned by neural activity which can be understood in causal terms—causes which should be identified and isolated—but that this view is unduly *reductive* of the organism (Potter & Mitchell, 2025, 1-3). In contrast, they argue for “non-reductive and temporally extended notions of causality” in which causality, fully understood, is a diachronic process that centers on the *meanings* of *neural states* (Potter & Mitchell, 2025, 4). They therefore seek a better understanding of cognitive and neural processes’ relationship (Potter & Mitchell, 2025, 4).

Strikingly, the authors purport to reconceptualize Descartes’ substance dualism by moving beyond synchronic and mechanistic causation: “Thoughts are not immaterial. They are *patterns of meaningful neural activity*” (Potter & Mitchell, 2025, 12).

⁹ Harman counter-argued: “What the hearer understands the speaker to say is, or ought to be, a function of what is said” (Harman, 1968, 232). Harman is effectively arguing that language use *is* a functional relation of speaker and hearer. Yet, the qualifier “or ought to be” is telling. Language use being stimulus-free means that one may not respond to a given stimulus at all. Such a problem does this create for explanations of it that when one does attempt to explain instances in which language is used, not only does the stimulus have to be tied to the utterance independently (from the scientist’s framework) of the utterance itself, but it would also have to account for why some stimuli do *not* cause language use—which requires the person to use language so that the stimulus can be identified!

The effort to link cognitive and neural processes in an explanatory framework is admirable. On reflection, however, their view would appear to merely restate the problem of ordinary language use and would re-assert neural activity as one of the *mechanisms* enabling it. (Note that Potter & Mitchell may not wish to apply their expanded view of causality to language use, though it is useful to ward off the very reductive concerns they hold by doing so.)

Following an example given by Potter & Mitchell (2025, 10-11), to explain language use in this sense, one would need to articulate the individual's internal representations at the moment of an utterance and elucidate how those representations are given meaning by the individual's history, distributed internal knowledge, and so forth.

That this process amounts to *explaining* language use in *causal* terms is unclear. For one thing, the analysis rests on both the individual speaker's and any hearer's interpretation of the *appropriateness* of the remark. This itself means that language processing in the speaker's and hearer's brain/mind is interacting with a host of other cognitive systems that allow these interpretations to be imposed. Yet, there is no apparent limit on the number of sentences within a given context that could count as appropriate nor any way to account in advance for the set of sentences deemed appropriate or inappropriate (see, McGilvray, 2009, 38-40). Thus, any attempt to account for actual, ordinary language use in this way would amount to a reflection of the individual's *own contribution*; a re-statement of the problem.

Moreover, stimulus-freedom remains a live problem. Ordinary language use is effectively *non-teleological*—there is no function for which the use of language has been pre-assigned nor can language's cognitive basis or neural properties account for its uses; there is no pre-defined set of uses (see section 7.3). By widening the notion of causality, it therefore loses its intended force: rather than well-defined notions of particular causes that reliably lead to particular outcomes, the search for patterns of meaningful neural activity invariably depends on the *individual using language*. While decidedly more refined, Potter & Mitchell's notion of causality may slip into "reasons" or "rational" explanations rather than "causes," the former belonging to the domain of common sense, the latter to scientific understanding (on this point, see, McGilvray, 2001, 13-18).

That the authors suggest thought is not immaterial because it instead refers to a pattern of neural activity merely reinforces the general point. This uses a theoretical framework to try and accommodate a seemingly unintelligible phenomenon—*thought*—through illustration. But illustration is not explanation.

The crucial insight arises here. The problem of establishing causality in either direction might lead one to throw up their hands and say, "*Of course* we cannot establish causality per se, but we—scholars—*know* that human language use is the result of a physical process! Why does CALU need to be such a pernicious problem as to expose the limitations of scientific inquiry into language and mind?"

The problem is that if ordinary language use is neither determined nor random yet appropriate, *then what is it?* It does little to suggest, with a gesture towards the unknown, that it could be

explained *in principle*—perhaps if we were equipped with different cognitive resources¹⁰—without simultaneously recognizing that no abstraction available to our inquiry appears adequate for handling the voluntary expression of thought.

Our scientific position on the matter depends on the abstractions we have available to us; the conceptual resources we can muster. If science is lacking, then our positions resort to *commonsense* explanations about why we say this or that. No account of the internal structure of language seems to provide any insight into *why* it behaves one way and not another; Turing’s contributions paved the way for a science of linguistic competence but a science of the full breadth of linguistic performance remains elusive. Human language use at least appears to be *intentional*. Yet, how do we intentionally ‘specify the supposed effect’ of our language use (Collins, 2006, 475-477)?

5. Large Language Models and Descartes’ Problem

Human language is thus a means of *at will* creation. It disturbs scientific inquiry today as it did for the Cartesians, pushing researchers to the limits of explanation. An ability at the heart of human intellectual endeavors undoubtedly has implications for AI.

It is little wonder, then, that the generative LLMs—capable of outputting grammatically and semantically coherent natural language text—appear to raise Descartes’ problem once more: how do we distinguish these machines whose outputs resemble our language use from actual humans?

The question, however, is not posed within the Cartesian framework. Thus, although Rees argues that GPT-3 “established a concept of language at the center of which does not need to be the human” (Rees, 2022, 179), his revival of Descartes’ problem neglects its reformulation. Computability theory never provided the tools to explain the *at will* expression of thought nor to construct a machine that reproduces this human ability. It allowed researchers to distinguish between “language” as an externalization (speech, the words on this page) and “language” as an internalized system of knowledge (capable of infinite generativity). Moreover, we make no distinction between the mechanical and the second substance; there are merely theories that deal with the problem of a system of unbounded linguistic generativity that is voluntarily and appropriately recruited.

We should therefore not *expect* generative LLMs to reproduce CALU. Descartes’ problem was not a relatively straightforward input-output test, despite its presentation in the literature (e.g., Riskin, 2016, 63; see also, Chomsky, 1980, 77-79). Moreover, and relatedly, algorithms are specified according to a finite set of instructions; the calculation proceeds in a finite sequence of steps that end (“halts”) with the answer; and the entire procedure is deterministic and fully specified (i.e., non-random) (Soare, 2016, 3). By design, this does not comport with what we have seen about CALU.

5.1 LLMs Do Not Reproduce CALU

¹⁰ The limits on human science do not imply limits in principle, but only those that have their roots in “our peculiar cognitive makeup” (McGinn, 1993, 87).

Still, a brief assessment is clarifying for two reasons. First, it allows us to gauge the future of AI via computation with confidence through the lens of Descartes' problem by determining where AI currently stands in relation to it. Second, moving from this, it allows us to take seriously the possibility that CALU represents a plausible candidate for the height which AI via computation cannot reach.

LLMs are neural networks comprised of artificial neurons usefully described as “computing units” (Ananthaswamy, 2024, 22). These units are organized in layers that possess adjustable parameters which denote the connection strength between units. LLMs' training procedure uses next-token prediction in which the model is given tokenized text¹¹ where the final token is ‘masked.’ The model then predicts what the masked token is and compares its prediction with the answer, as the training algorithm adjusts its parameters to improve its future predictions. Training continues until the model reliably predicts the masked tokens, thereby acquiring a statistical model of the training data (Ananthaswamy, 2024, 22-23).¹²

LLMs fare as follows:

Stimulus-Controlled: LLMs are controlled by internal and external stimuli in predictable ways and thus not stimulus-free. The output is inextricably tied to the input it receives (even if its operation over the input is separated by time; asynchronous). The LLM then operates over inputs according to the statistical model acquired during training. Unlike human language use, whether an LLM will generate an output can be predicted beforehand based on whether a value is being inputted into the model; a stimulus can be identified without post-hoc attribution.

Thus: (1) The LLM will generate an output value upon receiving an input value, unless instructed otherwise (merely reinforcing the point); (2) The LLM's outputs will inextricably be tied to the “environment” established by its internal programming and external prompting (e.g., the operation(s) it is to perform over the input value and the instructions provided by the prompter); and (3) The LLM will not alter the process by which it operates over an input unless it is provided with internal or external instructions to this effect.

Weakly Unbounded: LLMs can generate new sentences according to a particular configuration acquired via their training, which may be considered a form of weak unboundedness (see generally, McGilvray, 1999, 85-86). It is precisely this criterion that researchers *should* expect to find the most debate about LLMs as models of human linguistic competence, for the ability to generate (albeit on command) outputs judged grammatical by humans—and do so over an indefinite range (on command)—tilts closer toward familiar linguistic inquiry. And one does find this debate (e.g., Contreras Kallens et al., 2023; Katzir, 2023; Piantadosi, 2024).

On the strong interpretation, what matters is the model's ability to understand the *meanings* of these expressions and/or their *grammatical structure*. LLMs appear to be *weakly* unbounded; weak only relative to the full richness of human linguistic competence.

¹¹ This simply means that words are represented to the model as tokens rather than natural language characters.

¹² Models built on this architecture, including “reasoning” models with significant post-training innovations, do not fundamentally affect the argument here.

Descartes’ problem is about more than string-generation or the arrangement of tokens into a particular configuration; it is about the expression and complementarity of thought through language use. This stronger interpretation is about the infinite productivity of language use, the ability to generate form/meaning pairs in interaction with other cognitive systems over an unbounded range (see generally, Murphy & Leivada, 2022). The language faculty employs *contentless* generative procedures in concert with other systems to produce *contentful* expressions (Watumull, 2024, 12).

There are reasons to doubt LLMs achieve this strong unboundedness, including the persistence of hallucinations in their outputs; the excessive idealizing involved in their training procedure relative to human language acquisition (see, Dupre, 2024); failures to represent principled linguistic structure in frontier models like OpenAI’s o3 (Murphy et al., 2025a); and deficiencies in DALL-E 2 and DALL-E 3 on the visual representation of linguistic meaning relative to human children (Murphy et al., 2025b).

Functionally Appropriate: Being stimulus-controlled, LLMs exist in a *functional* relationship with their internal and external environments. More specifically, the alignment of the model with the expectations of the human end-user (through any technical means) restricts its linguistic output to serving a clearly defined goal in its environment. Any internal “model” or “conception” of “appropriate” output that exists within the LLM is premised on responding to human queries in useful and legal ways. Thus, the functional appropriateness of LLMs’ outputs is not *related* to internal or external stimuli (as it is with humans) but *caused* by them.

LLMs fail to reproduce CALU.¹³

6. CALU and the Upper Limit on Artificial Intelligence

That LLMs do not replicate CALU is not the most interesting aspect of a conversation on CALU and AI. Our focus turns instead to CALU as a strong candidate for a human ability not replicable through AI via computation. We do this through engagement with relevant scholarship on human characteristics relevant to any future artificial general intelligence (AGI) and determine whether their absence in today’s AI systems is what accounts for their failure to exhibit CALU.

6.1 Embodiedness and Agency

The original Cartesian language test conceives of the subject as what we might today call *embodied*. Those “machines bearing the image of our bodies” (Descartes, 2009/1637, 71) are closer to humanoid robots than LLMs accessed via a digital interface. This is not surprising—the Cartesians did not anticipate digital computers. The distinction between the mechanical body and non-mechanical soul was central (see, Riskin, 2016, 61-63). Although we operate in a different framework today, is an LLM’s *lack* of embodiedness responsible for its failure to reproduce CALU? Additionally, would some form of *agency*—often accompanying discussions of embodiedness—be sufficient for the artificial reproduction of CALU via computational means?

¹³ Note that even if LLMs possessed a strong unbounded capacity, this would not constitute CALU.

To the point, some scholars draw links between free will, agency, and embodiedness in conceptions of future AI. Mitchell advocates for embodiedness “either in physical robotics or in software entities that can act in simulated environments” (Mitchell, 2023, 297). This position flows from a dynamic view of biological organisms in which they use their capacities for creativity and open-ended thought in persistent interaction with their environments (Mitchell, 2023, 273-277).

For Mitchell, “*intelligence* and *life* may be inextricably intertwined, in ways that revolve round the concept of agency” (Mitchell, 2023, 295). AGI, Mitchell (2023, 296) argues, would require the ability to act appropriately in both new and uncertain situations by efficiently applying past knowledge and understanding to predict the consequences of possible actions. How does an agent accomplish this?

Causal intervention that allows agents to test causal relations in their environment that otherwise would not occur is central to this agency (Mitchell, 2023, 297). Such agents will also need a *reason* to act certain ways and thus require a “master function that can motivate action...Having goals gives actions value” (Mitchell, 2023, 297). Interestingly, Mitchell notes that “genuine agency” may require low-level *indeterminacy*, perhaps through “an added element of randomness or noise” (Mitchell, 2023, 298). The alternative is a purely deterministic machine that will “lack the flexibility and creativity needed to drive the kind of exploration of the world and of its own options that will allow it to accumulate causal understanding” (Mitchell, 2023, 299).

Mitchell’s view finds some convergence with work on *active inference*. Pezzulo et al. (2024, 102-104) argue that AGI depends on linking grounded interactions and sensorimotor predictions. In biological organisms, this requires embodiedness. The driving idea is that biological organisms must be sustainable. They therefore require *controlled interactions* with the world that allow the organism to reliably switch between states of affairs (e.g., eating to reduce hunger). Certain features of the environment are meaningful to organisms because they “specify the ways that we can act on the world” (Pezzulo et al., 2024, 98). Responding to these features is a “type of sensorimotor understanding” that *precedes* explicit knowledge of the world (Pezzulo et al., 2024, 98).

Enter active inference: predictions made by the organism about how actions will affect their internal states depend on “grounded world models” (Pezzulo et al., 2024, 98). Thus, unlike the bypassing of this process by LLMs, AGI requires grounded world models which in turn require embodiedness.

Yiu et al. (2024) report the results of experiments probing LLMs’ capacity for imitation versus innovation relative to human children. LLMs, they argue, are *cultural technologies*. They are distinguished from the *truth-seeking* epistemic functions of systems underpinning human perception, causal inference, or theory-formation (Yiu et al., 2024, 874-875). Most striking is their extended remark on the difference between LLM and human learning:

Although we do not know the details of children’s learning algorithms or data, we do know that, unlike large language and language-and-vision models, *children are curious*,

active, self-supervised, and intrinsically motivated. They are capable of extracting novel and abstract structures from the environment beyond statistical patterns, spontaneously making overhypotheses and generalizations, and applying these insights to new situations (Yiu et al., 2024, 880) (emphasis added).

Converging somewhat with Mitchell and Pezzulo et al., these authors emphasize that human children exhibit characteristics relevant to agency: they are self-directed, with their own internal motives, sustaining themselves in interaction with their environments in part through the novel and appropriate application of abstract structures. They notably go beyond active inference: human children do not merely exhibit these characteristics as a function of survival in a given environment; of state-changing necessities. Rather, they exhibit truth-seeking *curiosity* and *motivation* that may or may not have direct survival functions.

6.2 Would Embodiedness and Agency Yield CALU?

Several characteristics are posited above as central to agency, general intelligence, and the learning techniques required to achieve them: physical or simulated embodiedness and causal intervention therein, low-level indeterminacy, world models acquired through controlled interactions that are grounded in sensory modalities (which also entails embodiedness), and a bundle of truth-seeking functions including curiosity, activity, self-direction, and intrinsic motivation.

All these characteristics dance around, but never touch on, CALU. Its tripartite nature is distinct from these characteristics: it is not a capacity for causal intervention and the embodiedness this requires; it is not the building-up of world models grounded in sensorimotor predictions for the sake of reliable state-switching; though it is closer to curiosity, self-direction, and intrinsic motivation, these, too, are distinct from the problem of *at will* expression of thought.

It is conceivable that, by some definition, an AI-enabled system could exhibit these characteristics and be considered *intelligent* or in possession of *agency*. Causal intervention, robust understanding of the causal structures of scenarios, grounded world models, active learning, and so forth are meaningful contenders for achieving these. *The absence of CALU critically does not mean the absence of intelligence, agency, or the embodiedness either may require*. What is unclear is that these characteristics would be sufficient to yield the stimulus-free, unbounded, yet appropriate expression of internal content. This is unsurprising: the observations bound up in CALU grew out of a test not for intelligence, but for *mind*.

The foregoing remarks thus allow us to distinguish between agency, embodiedness, and the characteristics they either entail or yield (e.g., causal understanding) and CALU. They also highlight an assumption seemingly operative in some notions of agency: that humans are the prime movers of sufficiently advanced AI-enabled machines' sophisticated actions in the world. In particular, Mitchell's (2023, 297-299) tentative proposal to ensure that future systems exhibit low-level indeterminacy, perhaps through an element of randomness, aligns (perhaps incidentally) with Turing's (1950, 1951) emphasis on the *appearance* of intellectual freedom and the latter's own proposal for a random element. Notice that this reaffirms that the burden of prime movement or full autonomy is on humans who instill such randomness in machines.

Yet, it is the intellectual direction provided by the operator or the programmer noted by Jefferson (1949, 1109), with reference to Descartes, that constituted an argument for a limit on intelligent machinery. It is conceivable that these possible AI systems in possession of agency replicate a vast diversity of intellectual powers without ever being able to detach themselves, appropriately so, from the circumstances of their use over the indefinite range so enjoyed by humans; computational brains that lack creative minds, to speak entirely metaphorically.

6.3 Non-Teleological Language Use and the Limit on Computation

LLMs are stimulus-controlled. Their outputs are therefore functional or goal-oriented. Mitchell's (2023, 297-298) conception of agency is also goal-oriented in that it explicitly calls for a "master function" to be provided to the artificial agent so that its actions have value. Such a vision coincides with renewed interest in *teleology* in philosophy and science. Teleology can be broadly defined as the functions and goal-directedness of something owing to facts about its form (Kodaj et al., 2025, 1). Some aspects of biology may be usefully conceived as teleological. It is likewise plausible that some components of a future AGI, if constructed, will be teleological. But is human language use teleological?

To assess this, an argument by Fedorenko et al. (2024) is both representative and instructive for our purposes. The authors note that the study of language is divided between two hypotheses on its functions: one posits that language is designed to serve a *communicative function* in the transmission of knowledge, thoughts, and feelings. The other, that language *mediates* thought and cognition (Fedorenko et al., 2024, 575).

They argue that empirical evidence shows that language does not *necessarily* mediate thought (Fedorenko et al., 2024, 577-579), that the presence of intact language does *not* entail the presence of intact reasoning abilities (Fedorenko et al., 2024, 580), and that evidence from sounds, word forms, word meanings, syntax, and ambiguity all suggest that language is *for* efficient communication (Fedorenko et al., 2024, 580-582). "Language is unlikely to be a critical substrate for any form of thought" (Fedorenko et al., 2024, 582). Fedorenko et al. (2024, 583) (rightly) conclude their argument by noting the sheer transformational power of language-for-communication for human societies.

The framing of the argument immediately fails to make the necessary distinction between *competence* and *performance*. The result of not making this distinction is failing to delineate between language as a capacity of the human mind/brain and ordinary language *use*. They thus assume that the latter, by virtue of conflating it with the former, in fact has a *function* of some pre-defined kind (i.e., communication). The original and reformulated Cartesian problem is entirely absent from the discussion.

As Watumull (2024) vividly illustrates, language as an internal system "exists for its own sake" though crucially it can be used by humans to "[unify] the worlds of abstractions, minds, and bodies" (Watumull, 2024, 2). The language-for-communication thesis is deficient, attempting to reduce language use to serve as a tool for mere expressiveness (Watumull, 2024, 15-16). Yet, language is a "quite useless" tool:

Thus, whilst profoundly valuable as a cognitive tool for the art of self-knowledge, to say language is “for” anything other than “its own sake” strikes me as discordant. Like a work of art, the language faculty and the structures it creates simply exist, with no intrinsic utilitarian telos (Watumull, 2024, 20).

Language use serves no well-defined ends except those ends for which individuals recruit it.

Several points come into view. First, language use is not teleological. There is no meaningful sense in which the stimulus-free, unbounded, yet appropriate use of language to express the contents of one’s mind can be reduced to a single “function” such as communication. Second, this argument voids the scientific value of the term *communication* by stretching it beyond any reasonable limit. It neglects one of the most basic Cartesian insights: language use is stimulus-free; individuals may not use language *at all* or they may use it for *anything*. The language-for-communication thesis cannot account for this property.

More important is the implication for AI. That language use seems to not be *for* anything prior to the *person’s decision to use it* is bound up in the creative and intellectual potential of the human species. It is therefore relevant to the claim that CALU is the threshold which AI via computation is unlikely to meet.

Could an AI system be constructed via computation that achieves this non-teleological language use? Language use without pre-defined, pre-given, *stimulus-free* ends? Mitchell’s suggestion of building in low-level indeterminacy via randomness seems to offer one such answer, explicitly to *avoid* a wholly deterministic and non-creative machine. McGilvray similarly argued that achieving stimulus-freedom is simpler than achieving appropriateness, having to merely provide the machine with a built-in “randomizing device” so the machine’s outputs “have nothing to do with input or circumstance” (McGilvray, 1999, 85).

These suggestions are effectively Turing’s, in the sense of lowered ambition. They meet not the standard of stimulus-freedom but the *appearance* of stimulus-freedom where the stage has been set by humans. They reflect the lower ambition of Turing’s Imitation Game relative to the Cartesians’ language test. This is no slight against the idea of a random element in principle. Rather, it underestimates the severity of the Cartesians’ problem of ordinary human language use and the nature of the problem of at will expression of thought.

A device that has a random element might output grammatically or semantically coherent natural language text or sounds (LLMs’ stochastic outputs can readily be interpreted as such). But notice how this random element only counts as stimulus-free within the highly localized context of the test being conducted on it: the machine will generate a random, but coherent, natural language output. The humans will impose an interpretation of appropriateness on the outputs to make sense of them (an ability not in doubt).¹⁴ Yet, even if the human participants in the test are unaware that the programmers designed the system with a random element, the fact is the system was designed to function in this way. That those programmers can then isolate the machine in a

¹⁴ The human will then respond, and devices like LLMs—being sufficiently advanced in this respect—will output a response of sufficient relevance as to be deemed “appropriate” by the human. But the output is merely functional.

highly localized context, effectively binding the machine to a very specific set of stimuli, does nothing to achieve the stimulus-freedom enjoyed by humans in free-form, ordinary contexts.

The problem originally faced by Descartes was not the possibility of a mechanical being that could output words in an organized, controlled environment in which the humans—whose possession of minds is not in doubt—could engage in context-dependent conversation. Rather, it was the ability “to respond appropriately to the input while still being “detached” from that input (“stimulus-free”) [*sic*]” (Boeckx, 2010, 36). The Cartesians were ignorant of the possibility of the infinite generativity of a finite system. Turing’s and other luminaries’ innovation in the twentieth century was to show their ignorance and the obstacles it raised unnecessary. The Imitation Game, perhaps out of necessity to compel the building of intelligent machines without concern for intractable problems at mid-century (*a la* Gonçalves, 2023a; Abramson, 2011), addressed a different problem than Descartes: explanation was not the priority of Turing’s (1950) paper. An emphasis on the appearance of intellectual freedom over its actual possession thus took the place of the detection of a real-world property; of mind. A change in the character of the test for machine intelligence, the Imitation Game, therefore resulted in its reduced stringency and relevance to the original Cartesian problem. One can find (highly stage-set) experiments in which LLMs are claimed to have passed some version of Turing’s Imitation Game today (e.g., Jones & Bergen, 2025). Whatever the relevance of such results, they are not that of the Cartesian problem.

The computational specification of CALU is therefore likely beyond even the foreseeable horizon; AI since, and in part because of, Turing has lowered its ambition (perhaps a strategically wise decision).

7. Would We Know If CALU Were Achieved?

A final concern is that if a future AI system *were* to reproduce CALU—via computation or not—we would not be able to adequately test for this ability. Two doubts may accompany future data-interpretation: (1) Whatever contexts in which the system is observed have been effectively staged by humans, thereby making it stimulus-bound; and (2) There is no way to probe for the *actual* stimulus-freedom, unboundedness, or appropriateness of the system’s language use, as the observed examples could merely represent outputs that are afforded by a system that is more limited than it appears.

The concern has precedent outside of contemporary AI. Land (1974) argued that testing for stimulus-freedom is self-defeating: any deliberate attempt to draw out a reaction from the subject will amount to presenting the subject with particular stimuli and gauging the subject’s responses to them. But this, as Land (1974, 19-21) points out, is not only reducing the subject’s language use to a *function* of stimuli in the local environment, but it also places the tester in the position of adopting the notion of stimulus-control for which Chomsky so excoriated behaviorists!

We therefore have a problem. The test seemingly cannot be conducted in feasible ways. Moreover, any task-specific tests of current fascination would be insufficient, including ancient text translation, vaccine development, and general question-and-answering (see, Jones, 2024).

They do not target CALU's characteristics. The wisdom of Turing's lowered ambition in his Imitation Game is in evidence here.

We should be clear about what this problem is *not*. The act of testing non-human subjects for CALU may incorrectly lead one to conclude that the problem of CALU is no problem at all; an illusion. That there are limitations in testing for CALU, however, is not tantamount to there being no CALU at all. The problem exists as we find it among ourselves (humans); we cannot come at it from the "outside," so speak. This is why CALU disturbs our efforts at *explanation*—an explanation for this behavior *is* available in principle. That we lack this explanation owes to our intellectual limitations; we cannot come at it from the "right" angle.¹⁵

So, too, for tests designed to identify CALU in AI systems. Notice that this problem differs from those in typical performance measurements in AI, including the effort to achieve ever-higher scores on novel benchmarks. CALU a different kind of problem. The at will expression of thought over an unbounded range through natural language means that there will be no final test that decisively confirms the subject possesses a comparable intellectual freedom. No matter how one tests for it, *auxiliary evidence*—coming from other domains—is always in supply for the organism capable of this behavior. den Ouden explains:

The auxiliary evidence for the 'creative aspect of language use' is not final, nor can it be expected that it ever will be. If man is creative he will continue to respond to his creativity creatively. Thus to argue that there is a validating factor in the auxiliary evidence that has been explained and cited is not to contend that the case for human creativity in language use is closed and complete (den Ouden, 1975, 92-93).

No source of evidence of linguistic creativity could ever serve as the last word on the matter.

Thus, although constructing tests for CALU is a limited enterprise, the problem posed by it is no less real. It may ultimately turn out that an AI system that reproduces CALU will best evidence it by ignoring humans and going about its business, leaving humans not to perform controlled experiments (necessarily), but to merely observe them over time in the former's "natural" environments.

The upshot of the foregoing remarks is that it clarifies the *scope* of the competence-performance distinction in measuring machine capabilities. For example, Firestone (2020) invokes this distinction in both humans and machines. He distinguishes his interest in "humanlike processing" from the question of whether "such systems are truly "intelligent" or "minded" in some deeper sense, which can be asked independently of their similarity to humans" (Firestone, 2020, 26563). To be sure, this is accurate given the context. One *can* productively probe for underlying competencies by taking certain steps (e.g., placing humanlike constraints on machine or vice versa (Firestone, 2020, 26563; see also, Millièrè, 2024, 5)).

Yet, if CALU is the sum-total of human linguistic performance, and machines fail to exhibit this behavior, then there is a sense in which such machines simply lack linguistic performance. It is

¹⁵ Land (1974) notably does not take issue with the linguistic creativity observations themselves, but only the test to identify CALU.

not that their language use is different; it is that they do not *use* language at all. This does not rule out the competence-performance distinction in human-machine comparisons. It does, however, re-draw its scope.

8. Conclusion

The Cartesian problem of ordinary language use brings inquiry to the limits of explanation. To be sure, just as generative linguists could reformulate the problem along the lines of competence and performance in the twentieth century owing to computability theory (an unforeseen development), it is conceivable that another unforeseen innovation will one day bring the full problem to its knees. From the current vantage point, however, CALU is likely an unreachable threshold for AI via computation, thereby limiting the full scope of its creative and intellectual powers relative to humans.

There is a deeper reason to believe this: science has gradually moved further from “hard problems” like the voluntary expression of thought rather than closer via inroads in adjacent areas. Scientific theories offer indirect insight into phenomena. Humans lack direct access to the natural world, restricted to theoretical frameworks that are more or less suited to the task of explanation (see, Chomsky, 2009b). Potter & Mitchell’s (2025, 12) postulation that thoughts are not immaterial because they are, instead, patterns of meaningful neural activity does not resolve Descartes’ original concern but redefines “material” in a way that accommodates an otherwise unintelligible phenomenon—*thought*.

AI systems built in our image will foreseeably be restricted in a similar way. Just as we lack any direct way of conceptualizing the *will* underlying CALU, and likewise lack a means of specifying CALU computationally, such systems will likely be manifestations of what we can accommodate—computational brains of a metaphorical sort that never attain our creative minds.

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